1. Clinical Question & Core Hypothesis

Clinical Question

NMIBC tumors with molecular signs of high immunogenicity (high TMB and MSI/dMMR) nonetheless show variable clinical benefit from immunotherapies—why does neoantigen richness not always translate into effective local anti-tumor immunity? 2) Tier 1 paths used: I built on the Immunoscore/TIL axis by selecting a spatial immune-organization imaging feature (TLS-like lymphoid aggregates) and combined it with the two Tier-1 molecular immunogenicity anchors (MSI/dMMR and TMB) because these three elements form the mechanistic triad: neoantigen supply (TMB/MSI) → local immune organization (TLS/TIL) → effector function. 3) Tier 2 evidence supporting the claim: MSH6 (MMR component) supports MSI biology; liquid biopsy (ctDNA/CTCs) provides a way to measure dynamic TMB/MSI changes and detect emerging resistance; ER status, GARS1 and SARIFA provide orthogonal tumor-intrinsic or microenvironmental modifiers that could explain discordant outcomes. 4) Why I chose therapy_response as the endpoint: the central translational question is about variable benefit from immunotherapy in NMIBC (clinical decision-making and trial stratification), so a therapy-response proxy (anti–PD-1/PD-L1 or BCG response) is the most actionable endpoint. 5) What I deliberately EXCLUDED and why: I excluded broad/general 'biomarkers' and non-specific tumor-biomarker paths (Tier 3 generics) to keep the hypothesis mechanistically tight; I also kept Tier-2 modulators out of the core claim to respect the 1 imaging + max 2-anchor constraint and to preserve a clear causal question about how spatial immune structures gate neoantigen-driven responses?

Working Hypothesis

In NMIBC treated with anti-PD-1/PD-L1 therapy, TLS-like lymphoid aggregates are required to convert high neoantigen burden (as indicated by MSI/dMMR and high TMB) into an effective local anti-tumor T cell response. Specifically, tumors that are TLS-high and MSI/dMMR-positive or TMB-high will show higher objective therapy response rates than TLS-low counterparts with equivalent TMB/MSI status. Conversely, TLS-low tumors with high TMB or MSI/dMMR will exhibit limited responses due to immune exclusion and poor local priming. TLS-high tumors with low TMB/MSI may have intermediate responses, underscoring that TLS acts as a critical context for neoantigen-driven immunity. The primary testable prediction is a significant interaction between TLS status (imaging feature) and neoantigen burden (TMB and MSI/dMMR) on therapy_response in NMIBC patients receiving checkpoint blockade.

Element	Value
Imaging feature	TLS-like lymphoid aggregates
Anchor gene(s)	MSI/dMMR, TMB
Primary endpoint	therapy_response

2. Why This Gene — Not Others

Primary hypothesis gene(s)

MSI/dMMR: MSI/dMMR) nonetheless show variable clinical benefit from immunotherapies—why does neoantigen richness not always translate into effective local anti-tumor immunity? 2) Tier 1 paths used: I built on the Immunoscore/TIL axis by selecting a spatial immune-organization imaging feature (TLS-like lymphoid aggregates) and combined it with the two Tier-1 molecular immunogenicity anchors (MSI/dMMR and TMB) because these three elements form the mechanistic triad: neoantigen supply (TMB/MSI) → local immune organization (TLS/TIL) → effector function.

TMB: TMB and MSI/dMMR) nonetheless show variable clinical benefit from immunotherapies—why does neoantigen richness not always translate into effective local anti-tumor immunity? 2) Tier 1 paths used: I built on the Immunoscore/TIL axis by selecting a spatial immune-organization imaging feature (TLS-like lymphoid aggregates) and combined it with the two Tier-1 molecular immunogenicity anchors (MSI/dMMR and TMB) because these three elements form the mechanistic triad: neoantigen supply (TMB/MSI) → local immune organization (TLS/TIL) → effector function.

Alternative / secondary genes considered

Tier	Gene	Role / Reason excluded from core
Secondary	MSH6	Supporting modifier from knowledge graph
Secondary	GARS1	Supporting modifier from knowledge graph
Secondary	ctDNA/CTC	Supporting modifier from knowledge graph
Secondary	ER_status	Supporting modifier from knowledge graph
Secondary	SARIFA-positivity	Supporting modifier from knowledge graph

3. Clinically Testable Predictions

Prediction 1: retrospective/prospective NMIBC cohorts treated with anti-PD-1/PD-L1 with tissue available for TLS assessment by multiplex IHC/IF and for TMB/MSI testing (WES or targeted panels, MSI PCR/NGS);

Prediction 2: assessment of therapy_response (RECIST/irRECIST or clinical response criteria) as the primary endpoint;

Prediction 3: statistical interaction analysis between TLS status and TMB/MSI on response;

4. Minimum Viable Validation Plan

A. Retrospective Cohort Analysis

Data source: TCGA-BLCA (Firehose Legacy), ~400 patients - Cox PH model with interaction term for the imaging feature and anchor gene - Kaplan-Meier 2x2 stratification

B. Tissue-Level Validation

Methods: Multiplex IF / IHC / spatial transcriptomics

Key readouts: - Immune structure markers (e.g., CD20, CD3, PNAd) - Effector cell localization (CD8) relative to tumor nests - Anchor gene protein / phospho-protein expression - Stromal features (collagen density by SHG, if applicable)

Goal: Confirm that the proposed mechanism operates at the spatial / tissue level.

C. Functional Validation

Models: Ex vivo organotypic cultures, 3D co-culture, syngeneic mouse models

Perturbations: Genetic (KO/rescue) or pharmacologic modulation of anchor gene

Readouts: T-cell infiltration, cytotoxicity (granzyme B / IFN-gamma), tumor growth / survival

Goal: Establish causal relationship between anchor gene activity and immune efficacy.

5. Strengths, Caveats & Alternative Directions

Strengths

- Interaction-first biomarker strategy — asks WHY the imaging feature sometimes fails, not just whether it predicts outcome.

Caveats & Risks

- Hypothesis is computationally derived and requires experimental validation.

Alternative Directions

- MSH6: Could serve as an alternative mechanistic axis if the primary hypothesis is not supported.
- GARS1: Could serve as an alternative mechanistic axis if the primary hypothesis is not supported.